

PHYSIOLOGICAL RESPONSES TO AN ACUTE BOUT OF SPRINT INTERVAL CYCLING

ERIC C. FREESE, NICHOLAS H. GIST, AND KIRK J. CURETON

Metabolism and Body Composition Laboratory, Department of Kinesiology, University of Georgia, Athens, Georgia

ABSTRACT

Freese, EC, Gist, NH, and Cureton, KJ. Physiological responses to an acute bout of sprint interval cycling. *J Strength Cond Res* 27(10): 2768–2773, 2013—Sprint interval training has been shown to improve skeletal muscle oxidative capacity, $\dot{V}O_{2\max}$, and health outcomes. However, the acute physiological responses to 4–7 maximal effort intervals have not been determined. To determine the $\dot{V}O_2$, cardiorespiratory responses, and energy expenditure during an acute bout of sprint interval cycling (SIC), health, college-aged subjects, 6 men and 6 women, completed 2 SIC sessions with at least 7 days between trials. Sprint interval cycling was performed on a cycle ergometer and involved a 5-minute warm-up followed by four 30-second all-out sprints with 4-minute active recovery. Peak oxygen uptake ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) during the 4 sprints were 35.3 ± 8.2 , 38.8 ± 10.1 , 38.8 ± 10.6 , and 36.8 ± 9.3 , and peak heart rate ($\text{b} \cdot \text{min}^{-1}$) were 164 ± 17 , 172 ± 10 , 177 ± 12 , and 175 ± 22 . We conclude that an acute bout of SIC elicits submaximal $\dot{V}O_2$ and cardiorespiratory responses during each interval that are above 80% of estimated maximal values. Although the duration of exercise in SIC is very short, the high level of $\dot{V}O_2$ and cardiorespiratory responses are sufficient to potentially elicit adaptations to training associated with elevated aerobic energy demand.

KEY WORDS high-intensity intermittent exercise, energy expenditure, $\dot{V}O_2$

INTRODUCTION

Sprint interval cycling (SIC) is a form of low-volume high-intensity exercise that elicits skeletal muscle, cardiorespiratory, and metabolic adaptations similar to more prolonged, moderate-intensity aerobic exercise (5,21). Sprint interval cycling, modeled after the Wingate test, involves a series of all-out 30-second sprints against a predetermined flywheel resistance on a cycle ergometer, interspersed with 4-minute active recovery periods against no resistance (5,8,21). Chronic SIC increases

muscle oxidative capacity (5) and $\dot{V}O_{2\max}$ (7,14), enhances insulin sensitivity (1,21), reduces postprandial lipemia (8), and improves glucose stability in type 2 diabetics (13) with as little as 300 kJ of exercise per week (10,11). The high-intensity nature of SIC is thought to recruit all muscle fiber types and elicit an energy demand of approximately 250% $\dot{V}O_{2\max}$ (6). One session of SIC is less than 30 minutes in duration, although only 2–3 minutes involves actual work. Performing 3 SIC sessions per week involves considerably less time than the recommended U.S. public health guidelines (9) of accumulating at least $150 \text{ min} \cdot \text{wk}^{-1}$ of moderate-intensity physical activity or $75 \text{ min} \cdot \text{wk}^{-1}$ of vigorous-intensity physical activity. Although skeletal muscle metabolic responses to repeated all-out 30-second bouts of cycling with 4 minutes rest in between intervals has been reported (15,16,18,22) and whole-body metabolic responses to a single all-out 30-second bout of cycling (2) and to 2 successive all-out 30-second bouts of cycling separated by 4 minutes (3) have been studied, whole-body physiological responses to more than 2 successive all-out 30-second bouts of cycling have not been investigated.

Repeated all-out 30-second bouts of exercise separated by 4 minutes of rest depend increasingly on aerobic metabolism. This is, in part, because phosphocreatine (PCr) is nearly completely used in the first 10 seconds and glycogenolysis and glycolysis are markedly reduced after a first 30-second all-out bout (3,15,16,18,22). Bogdanis et al. (3) reported that the peak rate of $\dot{V}O_2$ was increased in a second 30-second all-out bout ($72 \pm 3\%$ $\dot{V}O_{2\max}$) compared with a first ($61 \pm 2\%$ $\dot{V}O_{2\max}$), with $49 \pm 2\%$ of the energy estimated to come from aerobic metabolism during the second bout compared with $34 \pm 2\%$ during the first bout. Whether the trend of increasing aerobic contribution continues with a greater number of bouts is unknown. Examining the $\dot{V}O_2$ changes during a series of repeated all-out sprints will aid in understanding the reported benefits of SIC on aerobic fitness and metabolic function.

Traditional, moderate-intensity continuous exercise programs are designed to elicit a target heart rate (64–76% heart rate max) or $\dot{V}O_2$ response (46–63% $\dot{V}O_{2\max}$). Sprint interval cycling elicits similar muscular, metabolic, and cardiovascular adaptations to training as traditional aerobic exercise (5,21,23), but the associated acute cardiorespiratory and metabolic responses during a training session have not been reported. The purpose of the current investigation was to examine the $\dot{V}O_2$ and cardiorespiratory responses during a series of four 30-second all-out sprints

Address correspondence to Eric C. Freese, efreese2@gmail.com.
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TABLE 1. Subject characteristics.*

Variable	Men (<i>n</i> = 6)	Women (<i>n</i> = 6)
Age, y	22.0 ± 3.2	20.8 ± 0.8
Height, cm	181 ± 5.1	166.6 ± 4.5
Mass, kg	80.77 ± 9.6	61.54 ± 8.8
%Fat	18.7 ± 4.3	27.8 ± 4.7
Fat-free mass	66.24 ± 8.49	44.92 ± 4.34

*Values are mean ± SD.

on a cycle ergometer. Understanding the $\dot{V}O_2$ and associated cardiorespiratory responses during SIC will provide further insight into this time-efficient training modality.

METHODS

Experimental Approach to the Problem

A descriptive research design was used for the study. Participants performed 2 acute bouts of SIC with at least 6 days between trials. During these sessions, $\dot{V}O_2$ and cardiorespiratory responses were measured to determine the strain on the cardiovascular and metabolic systems during this high-intensity intermittent exercise. Intraclass correlation coefficients (ICC) were calculated to assess the reliability of the outcome measures ($\dot{V}O_2$ ICC_{1,227} = 0.97; heart rate ICC (1,227) = 0.98; minute ventilation ICC (1,227) = 0.99). Data from the 2 trials were averaged to improve the stability

of the measures. Knowledge of the acute responses to SIC will provide insight into the metabolic adaptations that occur with chronic SIC.

Subjects

Twelve young recreationally active healthy adults, 6 men and 6 women, participated in this study. Exclusion criteria included participants who were smokers, had a body mass index greater than 30 kg·m⁻², or had a history of cardiovascular disease or diabetes (type I or II). Physical characteristics of the participants are shown in Table 1. The study protocol was approved by the institutional review board, and all subjects gave written informed consent after a description of the study procedures and associated risks.

Procedures

During the first visit to the laboratory, participants completed a medical history questionnaire to confirm eligibility. Body composition was then assessed using dual-energy x-ray absorptiometry (iDXA; GE Healthcare, Fairfield, CT, USA). The fat-free mass (FFM) was used to estimate the cycle resistance prescribed for the SIC exercise sessions. Participants were then familiarized to the protocol and performed 1 practice sprint. Over the next 2 weeks, participants performed 2 SIC sessions during which $\dot{V}O_2$ and associated cardiorespiratory responses were measured.

Sprint Interval Cycling

Before each SIC session, participants refrained from exercise and alcohol for 48 hours and no caffeine was consumed on the day of testing. Participants were instructed to hydrate by

drinking 2 glasses of water 2 hours before the SIC session. Participants arrived in the laboratory at 1900 hours. Height was measured on a wall-mounted stadiometer and weight was measured on a digital scale. Sprint interval cycling was administered on a mechanically braked stationary cycle ergometer (Monark Ergonomic 874E; Monark, GIM, Stockholm, Sweden) with resistance on the pedals set to 0.088 kp·kg⁻¹ FFM (8). Sprint interval cycling involved four 30-second all-out sprints with 4 minutes of active recovery between each sprint. Oxygen uptake was measured continuously during the last minute of the warm-up and during each sprint and recovery period by open-circuit spirometry using a PARVO Medics TrueOne

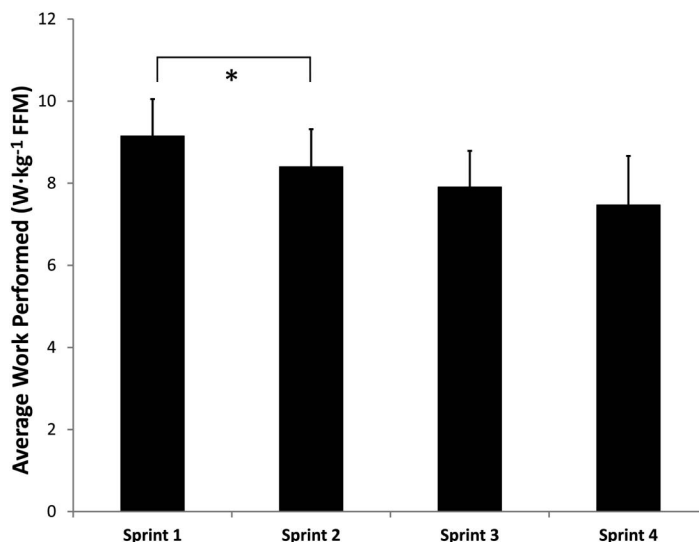


Figure 1. Average work performed (W·kg⁻¹ FFM) during 30-second sprints decreased across the duration of SIC (*p* < 0.05). Asterisk indicates statistically significant difference between sprints 1 and 2 (*p* < 0.05). FFM = fat-free mass; SIC = sprint interval cycling.

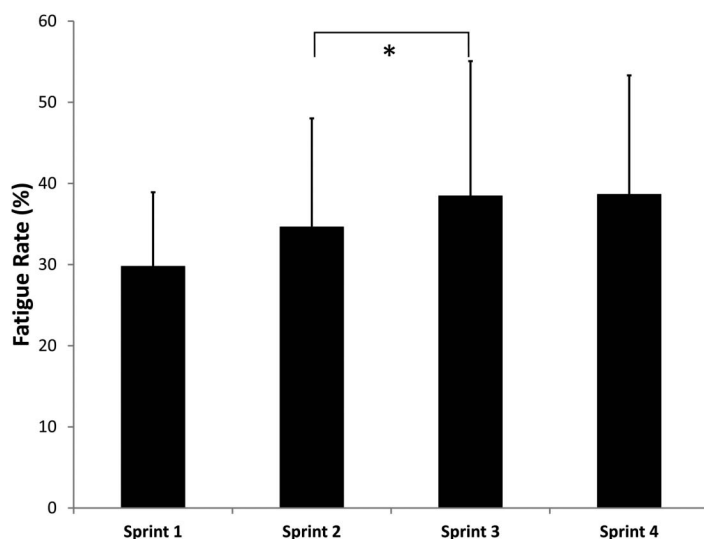


Figure 2. Fatigue rate (%) during 30-second sprints increased across the duration of sprint interval cycling ($p < 0.05$). Asterisk indicates statistically significant difference between sprints 2 and 3 ($p < 0.05$).

2400 Metabolic Measurement System (Parvo Medics, Inc., Salt Lake City, UT, USA). Heart rate was measured continuously with a Polar Vantage XL heart rate monitor (model 145900; Polar Electro, Inc., Woodbury, NY, USA). After a 5-minute warm-up pedaling against no resistance, participants were instructed to pedal as fast as possible, with resistance applied to the flywheel after maximal speed was attained and continued for 30 seconds. The number of revolutions and the total work performed on each sprint was quantified using an optical sensor (Sports Medicine Industries, Inc., St. Cloud, MN, USA) attached to the ergometer to measure flywheel revolutions. The sensor was interfaced to a computer

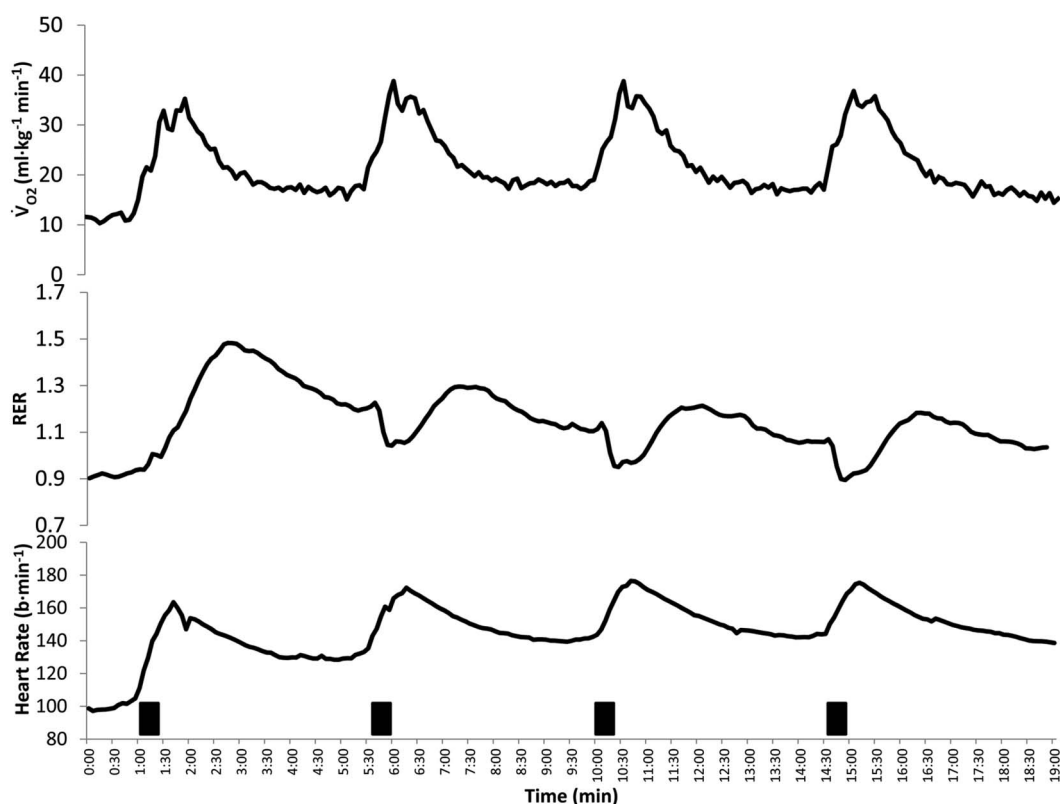


Figure 3. Oxygen uptake ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$), respiratory exchange ratio, and heart rate ($\text{b} \cdot \text{min}^{-1}$) during 4×30 -second bouts of "all-out" cycling (solid black bars) separated by 4-minute active recovery periods.



Figure 4. Minute ventilation ($\text{L} \cdot \text{min}^{-1}$) and oxygen pulse during 4×30 -second bouts of “all-out” cycling (solid black bars) separated by 4-minute active recovery periods.

and SMI Power software (version 1.02) recorded flywheel revolutions and calculated the power output each second. After each of the 4 sprints, participants actively cycled against no resistance for 4 minutes until the next sprint was started. This protocol was repeated until 4 sprints were completed. The time spent during cycling and recovery totaled 18 minutes.

Calculations

Oxygen uptake, ventilation, and heart rate values were averaged over 5-second periods. Oxygen pulse was calculated by dividing $\dot{V}\text{O}_2$ by heart rate. The work performed each second for the 30-second sprint was summed to obtain the work performed during each sprint. Fatigue was calculated as $([\text{Peak Power} - \text{Minimum Power}] / \text{Peak Power}) \times 100\%$.

Statistical Analyses

Statistical analyses were performed using SPSS for Windows (SPSS 18.0, Chicago, IL, USA). Peak responses were reported as mean $\pm$ SD. Unpaired *t*-tests were performed to determine differences between men and women in total work performed and fatigue rates during the sprints. A 1-way repeated-measures analysis of variance was conducted on work performed, fatigue rates, and 30-second average metabolic and cardiorespiratory responses across sprints. In the event of a significant

main effect, post hoc Bonferroni tests were performed to determine simple effects between consecutive sprints. An alpha level of 0.05 was used for all tests of significance.

RESULTS

The amount of work performed during each sprint was significantly higher ($p < 0.05$) in men (653.9 ± 88.5 W, 581.9 ± 116.6 W, 539.3 ± 132.9 W, and 520.4 ± 136.5 W) than in women (387.3 ± 45.2 W, 371.0 ± 49.8 W, 353.3 ± 49.6 W, and 352.4 ± 62.2 W). When expressed relative to FFM ($\text{W} \cdot \text{kg}^{-1}$ FFM), men performed more work in the first sprint (men: 9.7 ± 0.8 vs. women: 8.6 ± 0.3 ; $p < 0.05$) but performed the same amount of work as women in sprints 2–4 (men: 8.7 ± 0.9 , 8.0 ± 1.1 , and 7.7 ± 1.3 vs. women: 8.3 ± 0.6 , 7.9 ± 0.6 , and 7.8 ± 0.8). Average work performed decreased across the 4 sprints ($p < 0.05$) (Figure 1), and fatigue rates increased across the 4 sprints ($p < 0.05$) (Figure 2).

The $\dot{V}\text{O}_2$, respiratory exchange ratio, and heart rate responses during the last 1 minute of warm-up and the 18 minutes necessary to complete SIC can be found in Figure 3. Peak oxygen uptake ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) during the 4 sprints was 35.3 ± 8.2 , 38.8 ± 10.1 , 38.8 ± 10.6 , and 36.8 ± 9.3 , respectively. Oxygen uptake reached a peak 25 seconds after the first

sprint concluded, but reached a peak immediately at the end of sprints 2–4. Aerobic metabolism increased from sprint 1 to sprint 2 as evident by the increase in 30-second average $\dot{V}O_2$ ($24.87 \pm 9.09 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ vs. $30.21 \pm 9.10 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, respectively) ($p < 0.05$), but no further increase was evident in the third or fourth sprints ($30.94 \pm 9.06 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ and $30.53 \pm 8.31 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, respectively). The average $\dot{V}O_2$ response for 18 minutes of exercise was $22.4 \pm 8.4 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$. Peak heart rate ($\text{b} \cdot \text{min}^{-1}$) during the 4 sprints was 164 ± 17 , 172 ± 10 , 177 ± 12 , and 175 ± 22 . Heart rate reached a peak approximately 10–15 seconds after each of the sprints. Average heart rate response throughout the 18 minutes was $149 \pm 13 \text{ b} \cdot \text{min}^{-1}$. Minute ventilation (standard temperature pressure dry [STPD]) and oxygen pulse can be found in Figure 4. Peak minute ventilation ($\text{L} \cdot \text{min}^{-1}$) during the 4 sprints was 75 ± 36 , 102 ± 42 , 107 ± 42 , and 101 ± 36 and occurred at the conclusion of each 30-second sprint. Average minute ventilation throughout the 18 minutes of exercise was $61.6 \pm 23.5 \text{ L} \cdot \text{min}^{-1}$.

DISCUSSION

The primary findings of this study are that along with high assumed rates of anaerobic metabolism in the active skeletal muscles, relatively high levels of $\dot{V}O_2$, heart rate, and ventilation, averaging above 80% of estimated maximal values, are elicited for brief periods of time during and immediately after the bouts of cycling during a SIC session. The brief relatively high levels of $\dot{V}O_2$ and cardiorespiratory strain are of sufficient magnitude to potentially complement muscle adaptations in contributing to aerobic metabolic and cardiorespiratory adaptations after training.

During each 30-second sprint, a sharp rise in $\dot{V}O_2$ occurred, peaking at the end of the sprint. After a brief decrease, a secondary peak in $\dot{V}O_2$ occurred roughly 5–10 seconds after the sprint was completed. Heart rate and ventilation also increased during the 30-second sprint, but heart rate reached a peak 5–10 seconds after the sprint, although ventilation reached a peak at the end of each sprint. The secondary peak in $\dot{V}O_2$ occurred although ventilation and heart rate were decreasing. A secondary peak in oxygen pulse indicated that an increase in stroke volume or arteriovenous oxygen content difference contributed to the secondary peak in $\dot{V}O_2$. Reduced constriction on the vasculature within the working skeletal muscle may increase venous return and therefore stroke volume, causing a transient increase in cardiac output. In a similar study, the secondary peak was not observed after 3 bouts of 30-second isokinetic cycling exercise (17), possibly because of the different cycling modes (i.e., isokinetic vs. dynamic).

A limitation of this study was that maximal aerobic and cardiorespiratory responses to cycling were not measured. However, we have estimated the relative magnitude of the responses based on maximal values reported for untrained individuals of similar age. Age-estimated maximum heart rate (HR_{max}) ($220 - \text{age}$) is 198 and 199 $\text{b} \cdot \text{min}^{-1}$ for men and women in this study. Normative values for $\dot{V}O_{2\text{max}}$ of similar

age and fitness are 48.2 and 41.1 $\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, respectively (19). The average peak heart rate for each of the 4 sprints was $164 \pm 17 \text{ b} \cdot \text{min}^{-1}$, $172 \pm 10 \text{ b} \cdot \text{min}^{-1}$, $177 \pm 12 \text{ b} \cdot \text{min}^{-1}$, and $175 \pm 22 \text{ b} \cdot \text{min}^{-1}$, which ranged 83–89% estimated HR_{max} . The average $\dot{V}O_{2\text{peak}}$ during SIC was 40.7 and 38.4 $\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ for men and women, which is equivalent to 84 and 93% $\dot{V}O_{2\text{max}}$ age and sex norms. Although these estimates of relative intensity should be interpreted cautiously, the high percentages of estimated maximal values (9) indicate SIC elicits vigorous to near-maximal aerobic metabolic and cardiovascular responses for brief periods of time.

A significant increase in $\dot{V}O_{2\text{peak}}$ was evident from sprint 1 to sprint 2. Peak oxygen uptake did not increase further during sprint 3 or 4. The increase in fatigue and decrease in work done may have prevented further rise in $\dot{V}O_2$. Heart rate and ventilation continued to increase during the third sprint, indicating that cardiorespiratory strain continued to increase even without a further rise in $\dot{V}O_2$. Bogdanis et al. (3) found that anaerobic energy contributions decreased by approximately 41% during a second 30-second sprint, mainly because of an approximately 45% decrease in glycolysis. Although there was a decrease in anaerobic energy expenditure during the second sprint, total work done was only reduced by approximately 18% (3). The authors estimated that approximately 49% of the energy during a second sprint was derived from aerobic processes compared with approximately 34% during the first sprint. Using a similar protocol, Trump et al. (22) assessed PCr and other glycolytic intermediates before and after bouts 1 and 3 of a repeated isokinetic cycling protocol (30-second sprints, 100 rpm) with 4 minutes of rest between bouts. Their results suggest that PCr provides only approximately 15% of the total energy during the third sprint and that anaerobic glycogenolysis contributed approximately 10–15%, indicating that aerobic metabolism was the dominant energy producer during the third sprint bout. Thus, our results and those of others based on muscle enzyme changes suggest that after a second sprint, SIC is a predominantly aerobic activity. After SIC, $\dot{V}O_2$ remains elevated like it does after continuous endurance exercise (12). The high level of aerobic and cardiorespiratory responses during interval sprinting likely contribute to the reported increased $\dot{V}O_{2\text{max}}$ and perhaps other metabolic adaptations after SIC training (1,4,5,7,20,21).

As expected, average work performed decreased and fatigue rate increased across the 4 sprints. Total work performed decreased 8.2, 5.9, and 5.5% during the second, third, and fourth sprints, respectively. However, the decline in total work performed became progressively less across the 4 sprints and only the between sprint 1 and 2 was statistically significant. The progressively slower rate of decline over time was likely because of the reduced contribution of anaerobic metabolism and increased use of aerobic metabolism across the 4 sprints, although it also could have reflected “learning” to exert more relative effort. Bogdanis et al. (3) found that during repeated sprints, a 41% reduction in anaerobic energy supply during the second sprint bout was associated with only an 18% reduction in total work, suggesting an increasing importance of aerobic

energy systems during repeated sprint bouts. They estimated that aerobic metabolism provided approximately 49% of energy supply during the second sprint (3). Similarly, our finding of relatively little change in work rate across the 4 sprints, along with the increase in $\dot{V}O_2$, supports the suggestion that a majority of the energy expended during a SIC session was derived from aerobic metabolism.

We conclude that SIC elicits relatively high levels of $\dot{V}O_2$, heart rate, and ventilation, averaging above 80% of estimated maximal values, for brief periods of time during and immediately after the bouts of cycling during a SIC session. The brief relatively high levels of $\dot{V}O_2$ and cardiorespiratory strain are of sufficient magnitude to potentially complement muscle adaptations in contributing to aerobic metabolic and cardiorespiratory adaptations after training. The high levels of metabolic and cardiorespiratory processes elicited by an acute bout of SIC provide a foundation for understanding its positive effects on performance and health outcomes.

PRACTICAL APPLICATIONS

Our results highlight the importance of aerobic energy contributions during SIC. The strain placed on the cardiorespiratory and metabolic systems during SIC elicits near-maximal responses. The magnitude of the cardiorespiratory and metabolic responses complements previously reported muscular adaptations associated with high-intensity interval training that improve cardiovascular metabolic health and physical performance. Sprint interval cycling is a time-efficient model of exercise training that causes sufficient strain to improve cardiovascular and metabolic functions and physical performance. Time-efficient SIC may benefit those individuals for whom perceived lack of time is a barrier to exercise.

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